THESIS REVIEWER'S REPORT

I Identification data

Thesis title:	Distance estimate of autonomous helicopters by analysis of signal strength
Author:	Fedor Strjapunin
Type of thesis:	Bachelor
Faculty:	Faculty of Electrical Engineering (FEE)
Department:	Department of student
Thesis reviewer:	Ing. David Fiedler, Ph.D.
Reviewer's department:	Cornell University, School of Civil and Environmental Engineering

II Evaluation

Assignment	**Extraordinarily challenging**

The assignment seems demanding. The combination of signal analysis and reproducible research software development seems to be challenging. Each of those topics would be enough for a Bachelor thesis in my opinion.

Fullfilment of assignment	**Fulfilled with major objections**

The single main goal from the thesis assignment is "to analyse experimental data about location and signal strength from the flight of autonomous helicopters with the focus on reproducible research principles." In my oppinion, this goal has been achieved with the exception of the "reproducible research principles". From the four sub-goals listed in the assignment, all seems to be achieved, but all of them only partially. Therefore, despite I see the amount of work done as significant, I cannot say that the goals have been fully achieved.

Methodology	**Correct**

The methodology used for the signal analysis seems correct, which is supported by the results. However, the choices for the software implementation are questionable. One of the goals of the thesis is to "Discuss existing tools for reproducible research support and, if necessary, propose and implement a new system." I would inspect the Section 2.5 to provide a clear story that supports the necessity of creating a new system, and maybe the software choices, but there is none.

Technical level	C - Good

The thesis is technically sound, and the student seems to have a good grip of the topic of the ranging methods. However, there are some flaws in other sections: Section 2.5.2 use confusing terminology, and seems out of place in the thesis. The remaining sections (2.5.3 to 2.5.5) seems shallow and incomplete. Another flaw is the machine learning implementation. While the test data are held out for model training, they are used for hyperparameter (k) tuning, which makes the results even more questionable than the thesis admits. Finally, the limitations described in section 6.2.1 are quite significant, and their justification is not clear.

Formal and language level, scope	B - Very good

All the math and notation seems fine. The language is clear and understandable, with only a few minor typos. One thing I suggest for the future is reduce the use of passive voice. The only problem is the organization, where some of the paragraphs are not in the right place (e.g., the second half of Section 2.2 is clearly not a related work)

Sources and references	C - Good

Overall, the number and quality of references seems good. What I think is missing is rich integration of the references in the related work section. This section contains only a small number of references, and some claims are not supported by any reference (e.g., Section 2.1).

Additional comments

I would expect a thesis with the goal of reproducible research to be accompanied by a public repository for both the code and the data. Additionally, the images of the GUI in Section 6 are unreadable.

III Overall evaluation and questions

Considering the overall challenging goals, I grade the thesis **B - Very good**

Questions 1: Even after improvements achieved by this thesis, the abolute error seems stil to high (in meters), which is a limitation acknowledged by the in the conclusion. Do you see further improvement options with a potential to reduce the error at least by an order of magnitude? Or is it rather a fundamental limitation of RSSI, and a different approach should be considered?

Question 2: One of the goals of the thesis was to "following the reproducible research principles". However, the provided code targets a very small audience: Windows as the only supported platform, dependency on uv, or implementation as a VSCode plugin. As the logic behind these choices is not described in the thesis, I would like to ask for the strategy you have used to decide for these technologies that significantly limit the usability, and in consequence, the reproducibility of the research?

Date: June 1, 2026

Signature: